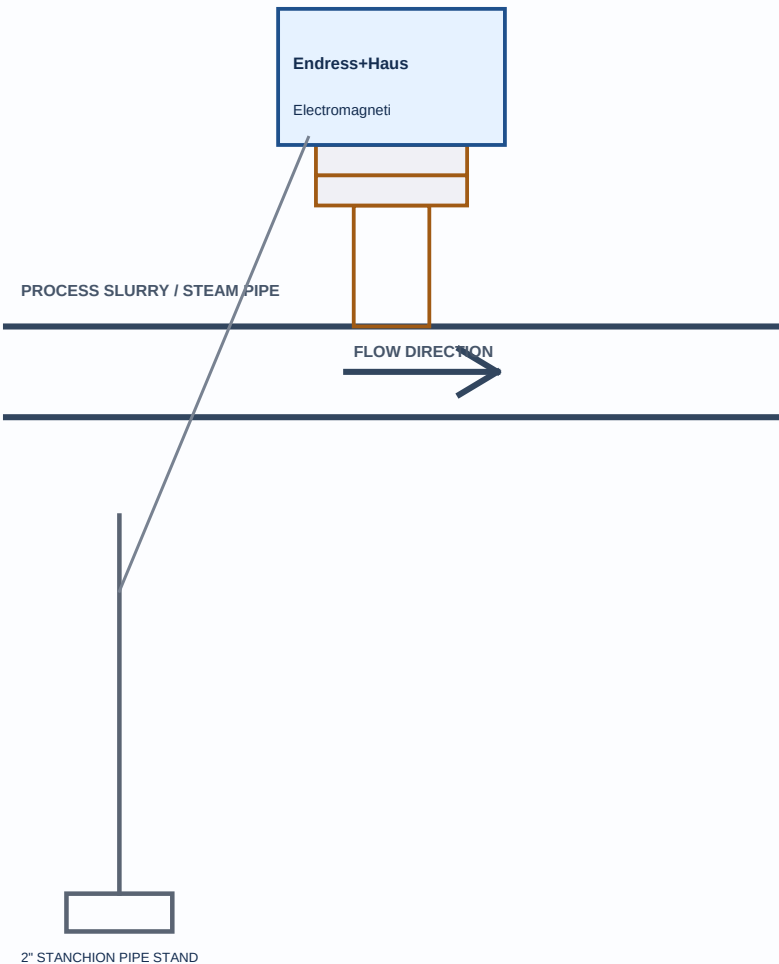


TEMPLATE TYP-07: ELECTROMAGNETIC FLOWMETER

TYPICAL TAGS: FT-40202 / FT-60201

SERVICE: Slurry & Process Water Volumetric Flow Measurement | PLC I/O: 1756-IF16 (4-20mA Flow) & 1756-IB32 (Pulse Counter)

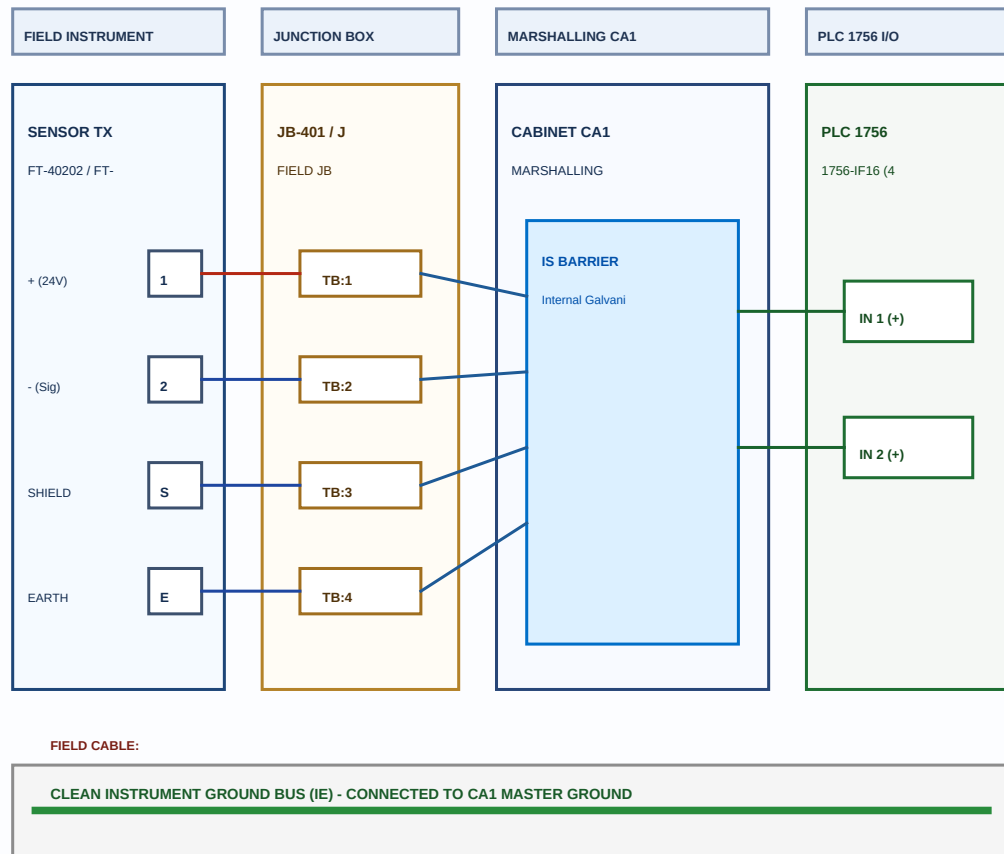
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN50/80/100 ASME CL150 RF F..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	NEMA 4X / IP67 Aluminum Tran..
Hazardous Class:	ATEX II 2G Ex eb Iia IIC T6....
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART (0-150 m³/h) + Scaled Pulses
Power Supply:	24V DC / 8W (4-Wire Electronics)
Field Cable Type:	1Px1.5mm² (Analog) + 1Px1.5mm² (Pulse/..
Destination JB:	JB-401 / JB-601
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	Internal Galvanic Isolation
PLC Module:	1756-IF16 (4-20mA Flow) & 1756-IB32 (P..

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Promag P 300 (5P3B50 / 80 / 1H PTFE Liner)

SERVICE: Slurry & Process Water Volumetric Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Promag P 300 Magmeter Sensor & E+H PTFE Lined, Hastelloy C Elec		1 EA
2	Grounding Rings / Discs SS316L E+H Grounding Electrodes Kit		2 EA
3	PTFE Envelope Flange Gaskets Chemical Resistant Gaskets		2 EA
4	Cable Glands M20x1.5 Ex d/e Nickel-plated Brass Glands		2 EA
5	Grounding Cables 6mm² Green/Ye Copper Earthing Strap with Lugs		2 EA

ENGINEERING INSTALLATION NOTES

- Straight piping requirements: min. 5D upstream and 2D downstream of flowmeter.
- Install grounding rings between meter flanges and pipe counter-flanges to ensure reference ground.
- Pipe must always remain 100% full of conductive liquid for accurate flow measurement.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONELECTROMAGNETIC FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-007
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
09 / 25